

The empirical closed form for OEIS A223828, proved

Adrian Perez Fontelles

Independent researcher

1 September 2026

Abstract

OEIS A223828 carries an empirical closed form for $a(n)$ — not a recurrence relating consecutive terms but an explicit formula. It is true, and it is decidable rather than empirical. The entry counts arrays subject to a condition on a bounded window of consecutive lines, so the lines of an array are the vertices of a finite digraph, an array is a walk in it, and $a(n)$ is a walk count on $S = 2752$ vertices. Any function of the form $\sum_j p_j(n) \lambda_j^n$ satisfies the monic linear recurrence whose characteristic polynomial is $\prod_j (x - \lambda_j)^{\deg p_j + 1}$; here that polynomial is $q(x) = (x - 1)^{21}$, of degree 21. Two things are then checked exactly: that the walk count satisfies the same recurrence from some index on, which the Cayley–Hamilton theorem reduces to a finite computation, and that the two sequences agree at 21 consecutive indices past that point. A monic recurrence determines every later term from its predecessors, so agreement there is agreement for good.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 The conjecture

OEIS A223828 is “Number of $n \times 5$ 0..2 arrays with rows and columns unimodal.”. It has offset 1 and begins

86, 7396, 310396, 7291180, 111026387, 1215505987, 10278415020, 70637615542, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = (63370093/405483668029440000) \cdot n^{20} +$
 $(71859847/9357315416064000) \cdot n^{19} + (33442223/166295420928000) \cdot n^{18} +$
 $(7404500641/2134124568576000) \cdot n^{17} + (27079914203/627683696640000) \cdot n^{16} +$
 $(25382116619/62768369664000) \cdot n^{15} + (28390454557/9656672256000) \cdot n^{14} +$
 $(289526123249/17118646272000) \cdot n^{13} + (3238876266163/41385738240000) \cdot n^{12} +$
 $(2848501575259/9656672256000) \cdot n^{11} + (17668229482801/19313344512000) \cdot n^{10} +$
 $(22667715892433/9656672256000) \cdot n^9 + (5949755731841/1188794880000) \cdot n^8 +$
 $(32017686101257/3621252096000) \cdot n^7 + (12478274965523/960740352000) \cdot n^6 +$
 $(13604598171413/871782912000) \cdot n^5 + (866466410064373/55576160640000) \cdot n^4 +$
 $(13639237471/1122750720) \cdot n^3 + (178704902203/22562971200) \cdot n^2 +$
 $(15738662/4849845) \cdot n + 1$

As of the “Last modified” line on the live entry (Oct 03 2025 22:41:04, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved. Write

$$\begin{aligned}
f(n) = & \frac{63370093n^{20}}{405483668029440000} + \frac{71859847n^{19}}{9357315416064000} \\
& + \frac{33442223n^{18}}{166295420928000} + \frac{7404500641n^{17}}{2134124568576000} \\
& + \frac{27079914203n^{16}}{627683696640000} + \frac{25382116619n^{15}}{62768369664000} \\
& + \frac{28390454557n^{14}}{9656672256000} + \frac{289526123249n^{13}}{17118646272000} \\
& + \frac{3238876266163n^{12}}{41385738240000} + \frac{2848501575259n^{11}}{9656672256000} \\
& + \frac{17668229482801n^{10}}{19313344512000} + \frac{22667715892433n^9}{9656672256000} \\
& + \frac{19313344512000}{5949755731841n^8} + \frac{32017686101257n^7}{1188794880000} \\
& + \frac{1188794880000}{3621252096000} + \frac{12478274965523n^6}{13604598171413n^5} \\
& + \frac{960740352000}{871782912000} + \frac{866466410064373n^4}{13639237471n^3} \\
& + \frac{55576160640000}{1122750720} + \frac{178704902203n^2}{15738662n} \\
& + \frac{22562971200}{4849845} + 1.
\end{aligned}$$

2 The count is a walk count

The entry’s defining condition is local: it is a condition on a bounded window of consecutive lines. Take as vertices the admissible configurations of such a window and put an edge from u to v when v may follow u , which the condition decides by inspection. An admissible array is then exactly a walk, so with M the adjacency matrix of that digraph on $S = 2752$ vertices, and u, w the vectors recording which windows may start and end an array,

$$a(n) = u^\top M^{g(n)} w$$

for the linear function g that the entry’s shape supplies. In particular a is C -finite: it satisfies some monic integer linear recurrence, though not necessarily a short one.

3 A closed form is a recurrence

Lemma 1. *Let $f(m) = \sum_j p_j(m) \lambda_j^m$ with the λ_j distinct and the p_j polynomials, and put $q(x) = \prod_j (x - \lambda_j)^{\deg p_j + 1}$. Then q applied to the shift operator annihilates f : writing $q(x) = x^r - \sum_{i=1}^r c_i x^{r-i}$,*

$$f(m) = \sum_{i=1}^r c_i f(m-i) \quad \text{for every } m.$$

Proof. The shift operator E acts on $m \mapsto \lambda^m p(m)$ as $(E - \lambda)(\lambda^m p(m)) = \lambda^{m+1}(p(m+1) - p(m))$, and $p(m+1) - p(m)$ has degree one less than p . So $(E - \lambda)^{\deg p + 1}$ kills $\lambda^m p(m)$, and the factors of q commute. $\square$

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A223828.
- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7; C -finite sequences, Section 4.1.)
- [3] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (Closed forms and linear recurrences with constant coefficients.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.